MIL-E-1/1219F(NAVY)
16 June 1975
SUPERSEDING
MIL-E-1/1219E(NAVY)
5 August 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 7AKP25

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRITPION: Cathode ray, electrostatic focus and deflection, rectangular, dual beam.

DIMENSIONS AND PIN CONNECTIONS: See figures 1 and 2.

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Esd	Eb3
Unit:	V	Vdc (Note 1)	Vdc	Vdc (Note 1)	Vdc	Vdc
Maximum:	6.9	0	2,750	4,950	5,200	10,000
Minimum:	5.7	-200	--- (Note 2)	3,250	3,000	6,500
TEST CONDITIONS (Note 4):	6.3	Adjust	Focus	Adjust	4,000	7,300

ABSOLUTE RATINGS:

Parameter:	ed	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	v	Meg	Meg (Note 3)	Vdc	Ratio	ft
Maximum:	770	1.5	3.0	±180	2.0	20,000
Minimum:	---	---	---	---	---	---
TEST CONDITIONS (Note 4):	---	---	---	---	---	---

GENERAL

Qualification - Required

Ⓕ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1						
5201	Electrode current (cathode)	-	Ib3 = 10 µAdc	Ik	---	100	µAdc
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	23	Ib3 = 25 µAdc	---	---	---	---
5101	Angle between traces	11		---	89	91	Degrees
5101	Angle between corresponding traces	12		---	---	---	---
(F) 5101	Trace and bulb alinement	13		---	88	92	Degrees
5106	Screen and faceplate blemishes	14		---	---	---	---
5223	Modulation	-	Ib3 = 2 µA	Ec	---	16	Vdc
---	Screen color	-	Orange	---	---	---	---
5226	Line width (electrostatic deflection)	15	Ib3 = 2 µAdc	---	---	0.50	mm
5231	Spot position (electrostatic deflection)	6,16		---	---	15	mm
5231	Spot displacement (leakage)	-		---	---	10	mm
5241	Grid-cutoff voltage	6		Eco	-50	-120	Vdc
5246	Focusing voltage at cutoff	6		Eb1	---	1,500	Vdc
5248	Deflection factor	6	1D2	DF	180	225	Vdc/in.
5248	Deflection factor	6	3D4	DF	140	170	Vdc/in.
(F) 5103	Pattern distortion	6,17		---	---	---	---
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No.1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5201	Electrode current (anode No. 1)	-	Ib3 = 25 µA	Ib1	-15	+10	µAdc
5101	Base alinement (electrostatic types)	-	+1D2, base key	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	+1D2	---	---	---	---
5101	Side terminal and base alinement	-	Base key	---	---	---	---
5101	Neck and bulb alinement (electrostatic types)	-		---	---	3.25	Inch

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Quality conformance inspection, part 2 -Continued						
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5101	Fact tilt	-		---	---	---	---
5216	Stray light emission	18		---	---	---	---
---	Screen brightness I	19,20		cBf	600	---	cB
---	Screen brightness II	19,20		D0.1 D1.0 D1.0-D10	400 325 50	--- --- 80	cB cB cB
5246	Focusing voltage	-	Ib3 = 25 uAdc	Eb1	1,150	---	Vdc
---	Tracking accuracy	6,21		---	---	2	%
5251	Heater-cathode leakage current	-		---	---	---	---
---	Post-accelerator lead pull test	22		---	---	---	---
(F) 1002	Barometric pressure, reduced	5		---	---	---	---
(F) 1331	Direct-interelectrode capacitance	-	Cathode to all Grid 1 to all D1 to D2 D3 to D4 D1 to all D2 to all D3 to all D4 to all	Ck Cg1 C1D2 C3D4 CD1 CD2 CD3 CD4	--- --- --- --- --- --- --- ---	7.0 8.0 6.0 6.0 11.5 11.5 11.5 11.5	pF pF pF pF pF pF pF pF
(F) 1141	Pressure (implosion)	-	30 psia	---	---	---	---
(F) 1006	Salt spray (corrosion)	-		---	---	---	---
(F) 5248	Deflection-factor uniformity	6		---	---	3	%
(F) ---	Interaction factor	7		---	---	$6x10^{-5}$	in./Vdc
(F) 5111	Vibration	-		---	---	1	mm
(F) ---	Humidity	8		---	---	---	---
(F) 1027	Temperature cycling (nonoperating)	-		---	10	---	Cycles
(F) 1026	Temperature range, operating (low)	9		---	---	---	---
(F) 1026	Temperature range, operating (high)	10		---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---
(F) 1105	Permanence of marking	-		---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	-	Group C	t	500	---	hrs
---	Life-test end points:						
5223	Modulation	-	Ib3 = 2 μAdc	Ec	---	18	Vdc
5226	Line width, gun A	-	Ib3 = 2 μAdc; positions X1, Y1, and Z1	---	---	0.55	mm
5226	Line width, gun B		Ib3 = 2 μAdc; positions P1, Q1, and R1	---	---	0.55	mm
---	Screen brightness I	-		cBf	550	---	cB

NOTES:

1. Accelerator power input (avg) should be limited to 6 watts. To avoid damage to the phosphor screen the exciting current density shall be not greater than 1.4 μA per cm^2.
2. Eb1 to Eb2 shall not exceed 3,200 V.
3. It is recommended that the deflection electrode circuit resistances be approximately equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.
4. All tests except vibration, interaction factor, and environmental tests shall be made on each unit separately.
5. The low-pressure test shall be made with maximum voltages applied, and pressure equivalent of 20,000 feet (348 mmHg). Connection shall be made to the base pins in a manner that does not degrade the tube breakdown characteristics. Satisfactory operation shall be the absence of acr-over and harmful corona.
6. The mean deflection plate and Eb2 potentials used for the line width tests shall be maintained throughout this test.
7. Deflection per unit volt of one beam at any position of the beam within its useful scan when a balanced voltage of at least ±300 volts is applied to the 1D2 deflection electrodes of the other electron gun shall be not greater than the value specified herein.
8. The humidity test shall be performed in accordance with the humidity test specified in MIL-T-5422. At the conclusion of the humidity test the post-accelerator lead pull test shall be performed.
9. Low temperature. Maximum allowable warmup time shall be 5 minutes.
10. High temperature. The temperature conditions of the test shall be the following:

    (a) Continuous operation at +85°C.
    (b) Intermittent or continuous operation for at least 30 minutes at +100°C.

11. Test for gun A made with 1.25 inch (31.75 mm) 3D4 trace centered at position Q of figure 5 and 4 inch (101.60 mm) 1D2 trace centered at G. Test for gun B with traces at positions Y and G. In addition, the change in orientation of the 1.25 inch (31.75 mm) 3D4 trace shall not exceed 2 degrees when deflected from Q to either P or R for gun A, and from Y to either X or Z for gun B.
12. Test shall be made using 4 inch (101.60 mm) 1D2 traces only centered at G of figure 5.
13. Test shall be made using a 4 inch (101.60 mm) 1D2 trace centered at G of figure 5. The angle made by the trace of each gun with respect to the shorter inside bulb wall shall be within the limits specified.
14. Phosphor blemishes shall be limited to maximum diameters of 0.030 inch (0.76 mm) with not more than 10 per tube, of which not more than 4 have diameters over 0.020 inch (0.51 mm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

NOTES: -Continued.

15. The line width test shall be performed as follows: Obtain a 45 to 55 line raster, 4 inches (101.50 mm) in the 1D2 direction by 1.5 inches (38.10 mm) in the 3D4 direction, oriented so that lines are generated in the 3D4 direction (vertical). With the mean 1D2 and 3D4 potentials at -155 V and -85 V, respectively, with respect to shield, adjust Eb2-to-shield potential within the range of 0 to -250 volts, for least astigmatism. Least astigmatism is secured when focus at the center occurs simultaneously in both the 1D2 and 3D4 directions (1D2 retrace and 3D4 trace directions). Adjust Eb1 for best overall focus, that is, for most nearly equal line widths at center and extremes of raster. Maintaining raster width at 1.5 inches (38.10 mm) and the mean 1D2, 3D4, and Eb2 potentials, use the shrinking raster method to determine line widths for gun A at P, Q, R of figure 5, and for gun B at X, Y, and Z. Two readings shall be obtained at each position, that is, one with raster lines vertical, and the other with raster lines horizontal. Line width at any position shall not exceed the value specified.

16. Spot position. The undeflected spot positions shall be within 15 mm squares whose centers are located respectively 0.500 inch (12.70 mm) vertically above and below the geometric center of the tube face or within a 15 mm square located at the center of the tube.

17. The edges of a 1-3/16 x 4-3/4 inch (30.18 x 120.65 mm) raster shall fall within the boundaries of the template as shown on figure 3. With the rasters from guns A and B displayed on the upper and lower screen areas respectively, as viewed from the front with anode terminal on the left, the template is positioned with the 0.125 inch (3.18 mm) dimension toward the outside. To minimize raster tilt effects, use 100 lines or more with high frequency on 1D2 plates.

18. The tube shall be operated for a minimum of 5 minutes with maximum potential (Esd=Eb2=4,950 V) applied to heater and accelerating anodes. Ec1 shall be maximum negative; positioning and deflecting voltages shall be zero. Viewed under dark conditions with the eye dark adapted (for 5 minutes), there shall be no evidence of fluorescent light when Eb1 is varied from minimum (800 volts) to maximum. This test shall be performed immediately following the completion of the other electrical tests.

19. Screen brightness I and II test conditions shall be as follows:

    Eb2=7.3 kV
    Esd=Eb2=4 kV
    Eb1=800 V
    Ib3=6.85 μA
    Raster size 73.5 x 73.5 mm

20. The phosphorescent light intensity observed t seconds after completion of the application of the nth raster to the tube is Dt(N), for example, D1(5) is the phosphorescent light intensity 1 second after the 5th raster. cBf shall be fluorescent light output under continuous raster excitation. The screen characteristics shall be measured under the following conditions:

    (a) Duration of raster = 1/60 second. The raster is formed by means of linear scanning with the horizontal scanning frequency 12 kHz and the vertical scanning frequency 60 Hz. The raster is repeated at 1 second intervals.
    (b) Distance of raster from axis of calibrated 931VA multiplier tube- 30 cm.
    (c) White light: Sufficient to de-excite the phosphor before buildup measurements so that the results are within 3 cB of those that would have been obtained after total de-excitation. After the tenth raster, excitation shall be removed. The light output 0.1 second (D0.1), 1 second (D1.0), and 10 seconds (D10) after excitation is removed shall satisfy the specified limits. Before performing the above measurement, illuminate screen with a 250 watt projection lamp for approximately 1 minute, then wait 30 seconds after lamp is turned off. A description of the primary standard replica lamps for the various screen types will be found in the Material Laboratory New York Naval Shipyard Final Report 5032-B15.40, dated 22 July 1959, copies of which may be obtained from the Naval Electronic Systems Command (code 5143). Centibels may be converted to milli-foot lamberts by the use of the following equation:

$$mftL = 3.88 \times 10^{-3} \times 10^{\frac{cB}{100}}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

NOTES: -Continued.

21. When 1-inch (25.40 mm) vertical traces are superimposed at the center of the tube and deflected horizontally ±2-1/2 (63.50 mm) inches by voltages proportional to the relative deflection factors, horizontal separation of corresponding points of the lines shall nowhere be greater than 2 percent of the deflection.

22. The anode cap bond shall be of sufficient strength to allow the tube to be supported by the anode lead without damage to the seal or condition of the potting compound.

23. This test shall be performed at the conclusion of the holding period.

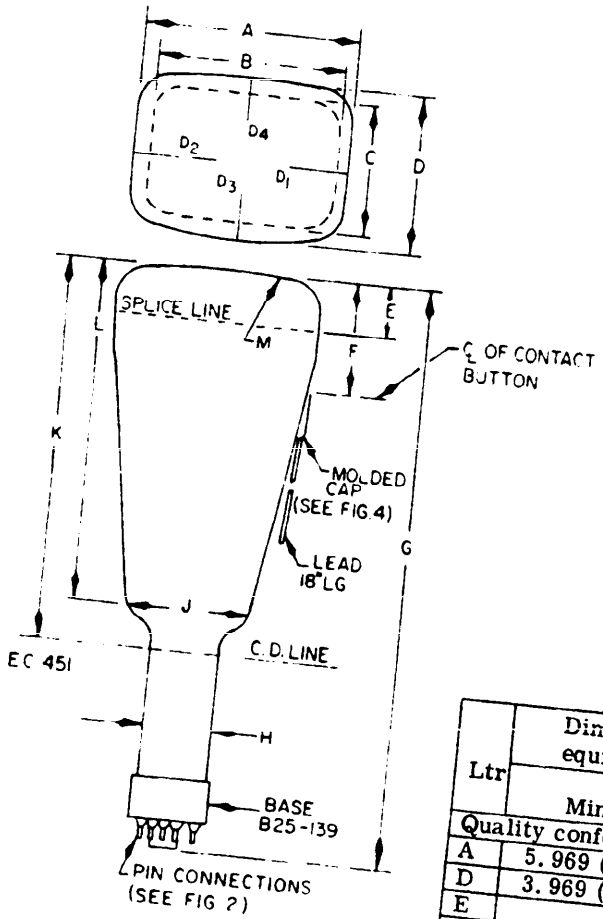
User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N035)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

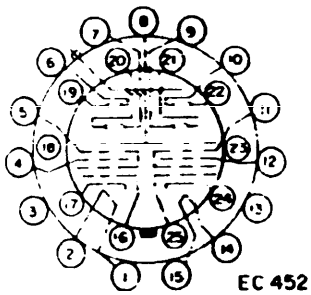


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.969 (151.61)	6.094 (154.79)
D	3.969 (100.81)	4.094 (103.99)
E		2.100 (53.34)
F	4.380 (111.25)	4.620 (117.35)
G	17.500 (444.50)	18.250 (463.55)
H	2.938 (74.63) D	3.062 (77.77) D
J		3.750 (95.25) D
K	10.375 (263.53)	11.250 (285.75)
L		10.600 (269.24)
Reference dimensions (see note)		
B	5.250 (133.35)	
C	3.250 (82.55)	
M	30.000 (762.00) R	

NOTE: Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 7AKP25.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

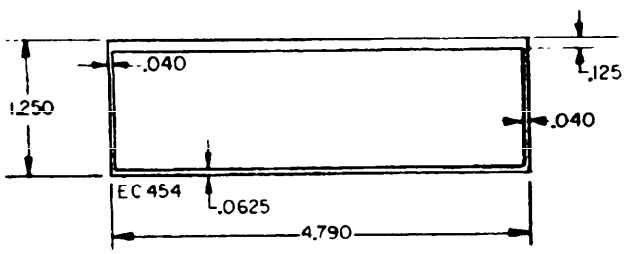


Bottom view

Unit A		Unit B	
Pin No.	Element	Pin No.	Element
14	Heater	1	Heater
15	Heater	2	Heater
24	Cathode	17	Cathode
25	Grid No. 1	16	Grid No. 1
12	Focusing electrode	4	Focusing electrode
22	Deflector D1	6	Deflector D1
10	Deflector D2	19	Deflector D2
9	Deflector D3	20	Deflector D3
21	Deflector D4	7	Deflector D4
11	Anode	5	Anode
13	No connection	3	No connection
23	No connection	18	No connection
		8	Shield

Cap-post accelerator

FIGURE 2. Pin connections for tube type 7AKP25.



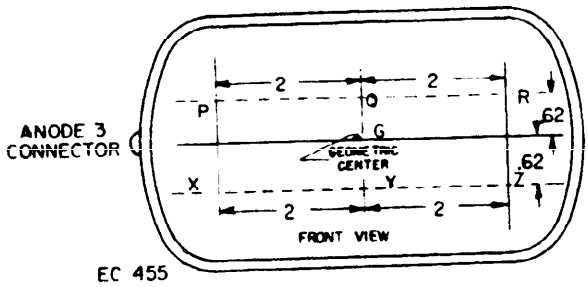
INCHES	MM
.040	1.02
.0625	1.59
.125	3.18
1.250	31.75
4.790	121.67

NOTE: Tolerance is ±.005 (.13 mm) for three place decimals.

FIGURE 3. Raster pattern template.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

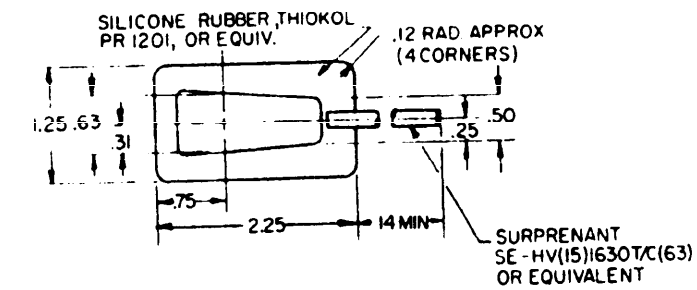
FIGURE 4. Molded cap for tube type 7AKP25.

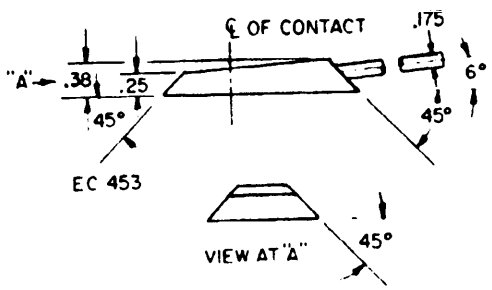


INCHES	MM
.62	15.75
2.00	50.80

NOTE: Tolerance is ±.02 (.51 mm) for two place decimals.

FIGURE 5. Line width test positions.





INCHES	MM
.12	3.05
.175	4.45
.25	6.35
.31	7.87
.38	9.65
.50	12.70
.75	19.05
1.25	31.75
2.25	57.15

Dimensions are maximum unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-01-29T21:46Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:16Z
Check the source to verify that this is the current version before use.